



SCHOOL OF COMPUTING SCIENCE & ENGINEERING

PROJECT APPROVAL FORM AND ABSTRACT

Odd 2025-2026

B. Tech Project Details

Project ID: -

Title	SNN-Based Malware Classification Using Program Execution Trace.		
Project Type	☐ Community based design problem (Interdisciplinary) ☐ App Development / Utility ☐ IOT/Hardware based ☒ AI/ML/Data Science ☐ Healthcare Projects	Project Outcome	☒ Project and Research Paper ☐ Project and Patent ☐ Project and Book Chapter
		SDG Goal Number	
Publication Target	☐ SCOPUS Journal ☐ SCOPUS Conference ☐ SCOPUS Book Chapter ☒ SCI Journal	Guide Name: Dr. Santosh Kumar	

Student Details:

S. No	Name	Enrollment Number	Admission Number	Program / Branch	Sem
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Sustainable Development Goals (SDG Goals), Targets and Indicators

- Goal 1: End poverty in all its forms everywhere
- Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture
- Goal 3: Ensure healthy lives and promote well-being for all at all ages
- Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
- Goal 5: Achieve gender equality and empower all women and girls
- Goal 6: Ensure availability and sustainable management of water and sanitation for all
- Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all
- Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
- Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
- Goal 10: Reduce inequality within and among countries
- Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable
- Goal 12: Ensure sustainable consumption and production patterns
- Goal 13: Take urgent action to combat climate change and its impacts
- Goal 14: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
- Goal 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
- Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
- Goal 17: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Blockchain-Secured Neuromorphic AI Provenance

Area/Domain of Project:

Cyber – Security & AI/ML/Data Science

Abstract:

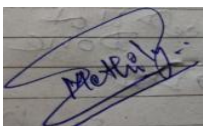
As advanced artificial intelligence, particularly brain-inspired neuromorphic models, becomes integrated into critical sectors like healthcare, finance, and autonomous systems, ensuring their trustworthiness is paramount. A significant vulnerability in the current AI lifecycle is the lack of a secure, transparent, and verifiable record of a model's origin and history—its provenance. Without this, AI models are susceptible to malicious tampering, data poisoning, or unauthorized modifications that can compromise their integrity and lead to catastrophic failures.

This research proposes a novel framework that leverages blockchain technology to establish an immutable and decentralized ledger for neuromorphic AI model provenance. In our system, every key event in a model's lifecycle—including the datasets used for training, architectural parameters, version updates, and deployment records—is cryptographically hashed and recorded as a permanent transaction on a blockchain. This approach creates a tamper-proof, time-stamped audit trail that can be independently verified. By securing the AI supply chain from creation to deployment, this framework ensures model authenticity, enhances transparency, and establishes clear accountability. The result is a foundational layer of trust, enabling the safe and reliable adoption of next-generation AI technologies in high-stakes environments.

Problem Statement:

The increasing deployment of complex artificial intelligence, particularly neuromorphic models, into critical infrastructure such as healthcare, finance, and autonomous systems introduces a significant and unresolved security challenge. The current AI lifecycle lacks a robust, transparent, and immutable system for tracking a model's provenance—its complete origin and history. This opacity creates a critical vulnerability across the entire AI supply chain.

Without a verifiable audit trail, AI models are exposed to a range of sophisticated threats that are difficult to detect. These include malicious tampering with model architecture, the injection of poisoned data during training to create hidden backdoors, and the unauthorized substitution of a validated model with a compromised one. The consequences of such integrity breaches are severe, ranging from flawed medical diagnoses and unfair financial decisions to catastrophic failures in safety-critical autonomous systems. The fundamental problem is the absence of a decentralized, tamper-proof framework to guarantee the authenticity and integrity of AI models, which erodes trust and poses a significant barrier to their safe and reliable adoption.



Signature of Students



Signature of Guide